Limits

Section 1: Algebraic Limits & Indeterminate Forms

1 Evaluate the limit: $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

a. 4

b. 8

c. 16

d. Does not exist

2 Evaluate the limit: $\lim_{x \to 0} \frac{\sqrt{1 + x} - 1}{x}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. 0

3 Evaluate: $\lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Section 2: One-Sided Limits & Continuity

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4 Evaluate the one-sided limit: $\lim_{x \to 3^+} \frac{|x-3|}{x-3}$.

a. 1

b. -1

c. 0

d. Does not exist

5 For what value of k is $f(x) = kx + 1$ for $x \leq 2$, and $f(x) = x^2 - k$ for $x > 2$ continuous at $x = 2$?

a. 1

b. $\frac{3}{2}$

c. 2

d. 3

6 What is $\lim_{x \to 0^-} \frac{1}{x}$?

a. $-\infty$

b. ∞

c. 0

d. Does not exist

Section 3: Limits at Infinity & Asymptotes

7 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{5x^3 - 2x + 7}{3x^3 + 4x^2 - 1}$.

a. $\frac{5}{3}$

b. 0

c. ∞

d. $\frac{7}{-1}$

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8 Evaluate: $\lim_{x\to\infty} \frac{2x+1}{\sqrt{9x^2+4}}$.

a. $\frac{2}{3}$

b. $\frac{2}{9}$

c. 0

d. ∞

9 Evaluate: $\lim_{x\to\infty} (\sqrt{x^2+4x} - x)$.

a. 2

b. 4

c. 0

d. ∞

Section 4: Trigonometric Limits & Definition of Derivative

10 Evaluate the special trigonometric limit: $\lim_{x\to 0} \frac{\sin(5x)}{2x}$.

a. $\frac{5}{2}$

b. $\frac{2}{5}$

c. 1

d. 0

11 Evaluate: $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

a. $\frac{1}{2}$

b. 1

c. 0

d. Does not exist

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12 The limit $\lim_{h \to 0} \frac{(2+h)^3 - 8}{h}$ represents the derivative of which function at which point?

a. $f(x) = x^3$ at $x = 2$

b. $f(x) = x^3$ at $x = 8$

c. $f(x) = 3x$ at $x = 2$

d. $f(x) = (x+2)^3$ at $x = 0$

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